Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Backend engineer with **2+ years of experience** building cloud-native, **multi-tenant** services, data pipelines, and performance-critical APIs across **AWS, Azure, Docker, and Kubernetes**, with strong ownership of reliability, debugging, and production operations.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Operating Systems, Software Engineering, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Architected a **multi-tenant AI platform** on Azure SQL, PostgreSQL, and Blob Storage for **500+ users**, reducing dashboard latency from **1.2s to 350ms** through schema design, indexing, and query optimization.
- Built **FastAPI**-based async orchestration and ETL-style data workflows across relational stores and cloud storage, processing **1K+ AI jobs** and **5K+ monthly integration jobs** with **99%+ reliability**.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Developed a **Flask** backend for a multimodal grading platform serving **150+ instructors** and handling **10K+ requests/day**, with secure ingestion, background execution, and cost-aware AI workflows.
- Implemented concurrent processing with **ThreadPoolExecutor**, **AWS RDS**, and **GitHub Actions CI/CD**, reducing feedback turnaround time by **60%** while maintaining stable API behavior under peak load.

USC AD Labs **Los Angeles**
ML Research Assistant *January 2025 – May 2025*
- Built simulation and model-training pipelines over **100K+ trajectory logs** and **10K autonomous-driving episodes**, preparing large-scale features for distributed-style experimentation in PyTorch and Hydra.
- Improved long-horizon planning by **18%** and closed-loop stability by **15%** through a spatial-temporal encoder and production-style evaluation loops for performance-sensitive systems.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered a cross-platform **microservice** data-sharing system with automated build, install, and validation pipelines across **Windows, Linux, and macOS** using Jenkins, PyTest, and Mockito.
- Optimized API and ETL workflows using **AWS Glue**, **S3**, and SQL, improving response times by **30%**, increasing pipeline efficiency by **68%**, and reducing technical debt by **20%**.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built Flask APIs, real-time sync, and webhook-driven payment workflows for a cross-platform booking platform supporting **50K+ records** with core API latency below **200ms**.
- Improved booking confirmation speed by **40%** and reduced manual refund handling by **70%** through Firebase-based event updates, payment automation, and analytics ingestion.

Skills

Languages: Python, C++, Java, SQL, JavaScript/TypeScript, Shell
Backend & Systems: FastAPI, Flask, REST APIs, Async Processing, Microservices, OOP, Distributed Systems
Cloud & Infra: AWS (RDS, S3, Glue), Azure, Docker, Kubernetes, PostgreSQL, MySQL, Linux
Reliability & Delivery: CI/CD, Jenkins, GitHub Actions, Testing Automation, Monitoring, Performance Optimization

Projects

FinNexus Live | *Next.js, Kubernetes, MongoDB*
- Built a scalable fintech platform for **50K+ users** and **1M+ transactions**, using **Kubernetes**, MongoDB, WebSockets, and CI/CD to support secure real-time transaction workflows at high volume.

Big-Brain Live | *FAISS, Pinecone, PostgreSQL, GCP*
- Engineered a cloud-based ingestion and indexing pipeline over **50K+ documents**, reducing retrieval latency by **40%** and enabling sub-second semantic search through optimized vector storage and retrieval flows.

Chat2SQL Repo | *Java, TypeScript, Docker*
- Extended the open-source Chat2DB project into an AI-powered SQL workspace supporting **10+ databases**, cross-platform local deployment, and structured query workflows across Java, TypeScript, Maven, and Docker services.